

Anomaly Detection Final Report

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Abstract—This final report presents a project of two unsupervised anomaly detection methods for credit card fraud detection under extreme class imbalance and limited label availability. Using the Credit Card Fraud Detection Dataset [1], we train both a Deep Autoencoder and a One-Class Support Vector Machine on legitimate transactions only. On the final test set, both models detect fraudulent behavior without supervised training; the Deep Autoencoder achieves stronger overall performance (Precision, F1, ROC-AUC, PR-AUC), while the One-Class SVM attains slightly higher recall with more false positives. The results show that the Deep Autoencoder provides a better balance between detecting fraud and keeping false alarms low.

I. INTRODUCTION

Credit card fraud continues to be a major problem for both the banking sector and consumers. The modern payment systems are built that they not only support high transaction but also facilitate quick-moving fraudsters. There are some difficulties recognizing fraud with respect to the unfavorable real-world data conditions: fraudulent transactions are very rare, labels for operation are delayed, and strategies for attacks evolve continuously over time. The said factors make it hard to develop systems for detection that are at the same time accurate and reliable in deployments.

The most important feature of the data from credit card transactions is the extremely imbalanced class distribution, where fraud is often represented by a very small number of records (usually less than 1%). In these situations, the normal supervised classifiers may reach misleadingly high accuracy rates. Besides, the supervision method relies heavily on top-quality labels for the cases of fraud. On the other hand, fraud patterns are subject to concept drift: fraudsters can have different tactics, which means that not every fraudulent action has the same structure. So, a model should be designed to identify all fraud patterns.

Considering these restrictions, the project concentrates on using unsupervised anomaly detection as the least labelling dependent option for spotting fraud. Rather than seeking to establish a direct link between features and fraud labels, unsupervised methods are targeting to understand the profile of legitimate transactions and accordingly mark deviations as potential frauds. This view closely resembles the day-to-day operation: the number of legitis is high and their verifications are quick, while on the opposite side, fraud labels are rare and inconsistent. On the other hand, unsupervised fraud detection presents a huge number of challenges, such as threshold selection, false positives management in the bulk of transactions, and anomaly detection that are not merely the result of preprocessing or feature distributions.

In the present final report, we devise and analyze two unsupervised methodologies for detecting credit card fraud: a Deep Autoencoder (AE) and a One-Class Support Vector Machine (OC-SVM). By conducting an evaluation of both techniques with the

same data preprocessing steps and experimental setup, we are able to offer a systematic comparison of their respective real-world advantages and disadvantages, especially regarding the detection of rare classes.

II. RELATED WORK

A. Background and Early Studies

Early attempts to detect credit card fraud using machine learning focused on classical neural networks and probabilistic modeling. Aleskerov et al. [2] introduced one of the earliest intelligent fraud detection systems by utilizing self-organizing maps and adaptive network structures. Srivastava et al. [3] later applied Hidden Markov Models (HMMs) to model sequential transaction behavior, demonstrating the value of temporal pattern analysis for fraud detection. Although pioneering, these methods face difficulties in scalability and adaptation to high-dimensional, anonymized datasets such as the European 2013 credit card dataset used in our project.

B. Unsupervised Anomaly Detection

With the limitations of supervised learning, unsupervised models have gained considerable attention. Rezapour [4] compared several anomaly detection techniques—including autoencoders, One-Class SVM, and Mahalanobis distance—and found that autoencoders achieved superior reconstruction-based discrimination between normal and fraudulent transactions. Zhang et al. [5] designed optimized deep learning frameworks for fraud detection and demonstrated that autoencoder-based latent representations significantly improve precision and recall when properly regularized. The One-Class SVM has also been widely used as a benchmark model for anomaly detection. Studies show that when trained exclusively on normal samples, OC-SVM can effectively identify novel fraudulent behaviors during deployment [6], [7].

C. Recent Advances Beyond Autoencoders and OC-SVM

To align with the instructor's requirement for reviewing recent studies even outside our chosen method, we examined additional modern techniques: Graph Neural Networks (GNNs) have demonstrated strong performance in transaction networks, where each transaction is a node and edges represent relationships; notable examples include GNN-based fraud detectors that exploit relational structures [8]. Generative Adversarial Networks (GANs) have been used to synthesize realistic fraud-like patterns and improve minority-class representation [9]. Transformers and attention-based models have been applied to financial fraud detection, capturing long-range dependencies across transaction sequences

[10]. Contrastive and self-supervised learning approaches have recently been shown to enhance anomaly detection by learning better representations without labels [11]. These works highlight the growing research interest in more expressive models, while also validating the relevance of autoencoders and OC-SVM as strong unsupervised baselines. The literature supports our methodology selection by demonstrating both the effectiveness and continued use of these models in the fraud detection community.

III. PROPOSED APPROACH

A. Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote the transaction dataset, where $x_i \in \mathbb{R}^d$ is a feature vector and $y_i \in \{0, 1\}$ indicates whether the transaction is legitimate (0) or fraudulent (1). In our setting, labels are used only for evaluation; model training follows an *unsupervised anomaly detection* paradigm.

The objective is to learn an anomaly scoring function $s(x)$ such that fraudulent transactions receive higher scores than normal transactions in expectation:

$$\mathbb{E}[s(x) \mid y = 1] > \mathbb{E}[s(x) \mid y = 0]. \quad (1)$$

We compare two unsupervised methods: (i) a reconstruction-based Deep Autoencoder (AE) and (ii) a boundary-based One-Class Support Vector Machine (OC-SVM). We select AE and OC-SVM as representative reconstruction-based and boundary-based unsupervised methods, respectively, enabling a systematic comparison of two distinct anomaly detection principles.

B. Pipeline Overview

Figure 1 summarizes the complete workflow used in our Colab implementation.

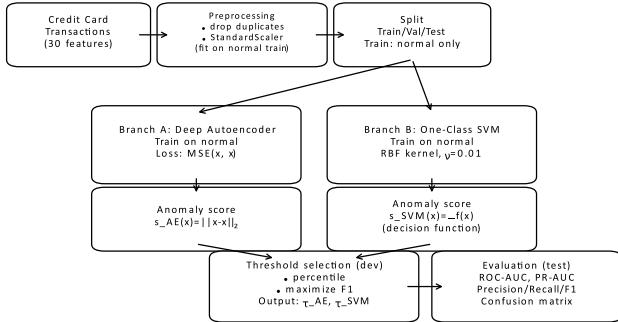


Fig. 1. Overview of the proposed unsupervised fraud detection pipeline used in our experiments. Both models are trained on normal transactions only. Anomaly scores are computed on dev/test splits and thresholds are selected using the dev split.

As illustrated in Fig. 1, both models share identical preprocessing, split strategy, and evaluation protocol. The pipeline consists of: (1) preprocessing and splitting, (2) training on normal-only data, (3) computing anomaly scores on dev/test splits, (4) selecting decision thresholds using the dev split, and (5) evaluating using metrics suitable for extreme class imbalance.

C. Dataset, Preprocessing, and Split Strategy

We use the *European Credit Card Transactions* dataset (2013), containing anonymized PCA-transformed features ($V_1, \dots, V_{28}$) and raw attributes `Time` and `Amount`, resulting in $d = 30$ numerical input features.

Since both models are sensitive to feature scale, we apply standardization:

$$x' = \frac{x - \mu}{\sigma}, \quad (2)$$

where μ and σ are computed *only* from the **normal** training data to prevent data leakage. Duplicate rows are removed prior to splitting.

To match the unsupervised setting, we train using normal samples only. We form:

- **Train + Validation (Dev):** legitimate transactions only.
- **Test:** contains both legitimate and fraudulent transactions for unbiased evaluation.

Threshold selection is performed on the dev split and final reporting is done on the held-out test split, ensuring the threshold is not tuned on test data.

D. Deep Autoencoder for Reconstruction-Based Anomaly Detection

1) *Architecture:* The autoencoder consists of an encoder $f_\theta(\cdot)$ and decoder $g_\phi(\cdot)$:

$$z = f_\theta(x), \quad \hat{x} = g_\phi(z). \quad (3)$$

In our implementation, $d = 30$ and we use a symmetric bottleneck architecture:

$$30 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 30, \quad (4)$$

where the middle layer of size 4 is the latent bottleneck. Figure 2 illustrates this design.

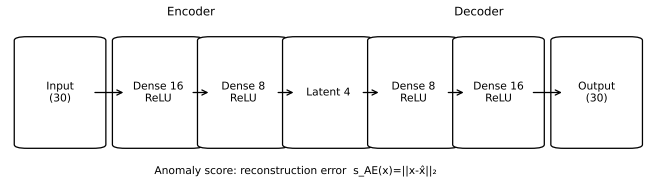


Fig. 2. Deep autoencoder architecture used in Colab. The anomaly score is derived from reconstruction error.

2) *Training Objective:* The model is trained on normal transactions by minimizing the mean squared reconstruction loss:

$$\mathcal{L}_{AE}(x) = \|x - \hat{x}\|_2^2. \quad (5)$$

We use Adam optimization (learning rate 10^{-3}) with early stopping and learning-rate reduction on plateau to stabilize training.

3) *Anomaly Scoring and Decision Rule:* After training, the anomaly score is defined via reconstruction error:

$$s_{AE}(x) = \|x - \hat{x}\|_2. \quad (6)$$

A transaction is predicted as fraud if:

$$\hat{y} = \begin{cases} 1, & s_{AE}(x) > \tau_{AE}, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

E. One-Class SVM for Boundary-Based Anomaly Detection

1) *Model Definition*: OC-SVM learns a boundary around normal data in a feature space induced by a kernel $\phi(\cdot)$. The primal optimization objective is:

$$\min_{w, \rho, \xi} \frac{1}{2} \|w\|^2 + \frac{1}{\nu N} \sum_{i=1}^N \xi_i - \rho \quad \text{s.t.} \quad w^\top \phi(x_i) \geq \rho - \xi_i, \quad \xi_i \geq 0. \quad (8)$$

We use an RBF kernel and set $\nu = 0.01$ in Colab, reflecting the assumption that anomalies are rare.

2) *Geometric Interpretation (Diagram)*: Unlike an autoencoder, OC-SVM does not have a layer-by-layer architecture. Its core behavior is geometric: it encloses normal data within a learned region and flags points outside as anomalies. Figure 3 visualizes this concept.

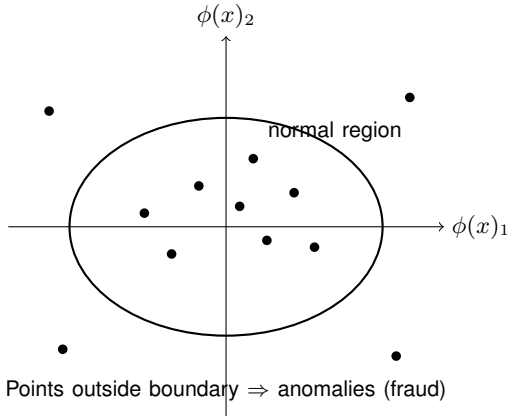


Fig. 3. Geometric interpretation of OC-SVM in feature space $\phi(x)$. The model learns a boundary enclosing normal transactions; points outside are considered anomalous.

3) *Anomaly Scoring and Decision Rule*: OC-SVM produces a decision function $f(x)$; points outside the learned region have lower values. We convert it into an anomaly score such that larger values indicate higher anomaly likelihood:

$$s_{SVM}(x) = -f(x). \quad (9)$$

Classification is obtained using a threshold:

$$\hat{y} = \begin{cases} 1, & s_{SVM}(x) > \tau_{SVM}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

F. Threshold Selection and Evaluation Metrics

A key challenge in unsupervised fraud detection is selecting τ to balance false alarms and missed fraud. In our experiments, thresholds are selected on the dev split using:

- percentile-based thresholds (e.g., 95th, 99th percentiles of scores),
- and threshold sweeping to maximize F1-score on the dev split.

On the final test split, we report both threshold-dependent and threshold-independent metrics:

- Precision, recall, and F1-score at the chosen threshold,
- ROC-AUC using continuous anomaly scores $s(x)$,
- PR-AUC (Average Precision) using continuous anomaly scores $s(x)$, which is more informative under extreme imbalance.

This unified protocol enables a fair comparison between reconstruction-based (AE) and boundary-based (OC-SVM) anomaly detection under identical preprocessing and split conditions.

IV. RESULTS AND DISCUSSION

This section reports the experimental results of the Deep Autoencoder (AE) and One-Class SVM (OC-SVM) under the unified unsupervised evaluation protocol described in Section V. All performance metrics are computed on the held-out **final test set**, while threshold selection is conducted exclusively on the dev split to prevent data leakage.

A. Dataset Characteristics

After preprocessing, scaling, and duplicate removal, the dataset is split as follows:

- **Training (normal only)**: 158,621 samples
- **Validation (normal only)**: 39,656 samples
- **Test-Dev**: 42,724 samples (42,488 normal, 236 fraud)
- **Test-Final**: 42,725 samples (42,488 normal, 237 fraud)

Fraudulent transactions represent approximately 0.55% of the test data, highlighting the severe class imbalance. The sample distribution across splits is visualized in Appendix Fig. 8.

B. Deep Autoencoder Results

The Deep Autoencoder demonstrates strong separation between normal and fraudulent transactions. On the dev set, fraudulent transactions exhibit a substantially higher mean reconstruction error (20.59) compared to normal transactions (0.81), yielding a separation ratio of approximately $25.4\times$. The distribution of reconstruction errors is illustrated in Fig. 4.

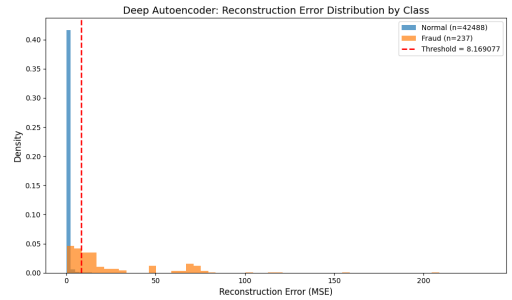


Fig. 4. Reconstruction error distributions for the Deep Autoencoder on the dev set (normal vs. fraud).

Threshold selection using F1-score maximization on the dev set yields $\tau_{AE} = 8.17$. Applying this threshold to the final test set results in:

- Precision: 0.449 (Appendix Fig. 10)
- Recall: 0.650 (Appendix Fig. 10)
- F1-score: 0.531

The autoencoder correctly detects 154 out of 237 fraudulent transactions, while producing 189 false positives. This corresponds to a false positive rate of 0.44% and a specificity of 0.996. The confusion matrix for the AE model is shown in Fig. 5.

Threshold-independent evaluation confirms strong ranking performance:

- ROC-AUC: 0.961 (Appendix Fig. 9)
- PR-AUC: 0.440 (Appendix Fig. 10)

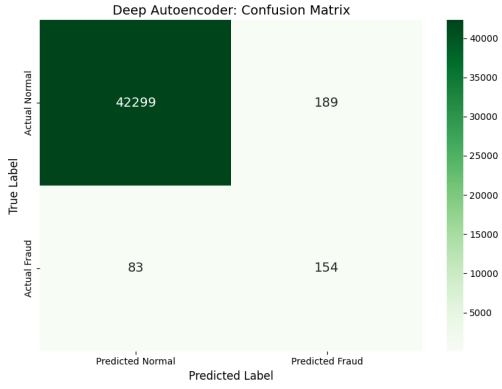


Fig. 5. Confusion matrix of the Deep Autoencoder on the final test set using τ_{AE} .

C. One-Class SVM Results

The One-Class SVM also provides meaningful anomaly separation. On the dev set, fraudulent transactions have a higher mean anomaly score (1.40) than normal transactions (-2.83). The anomaly score distributions are shown in Fig. 6. Using

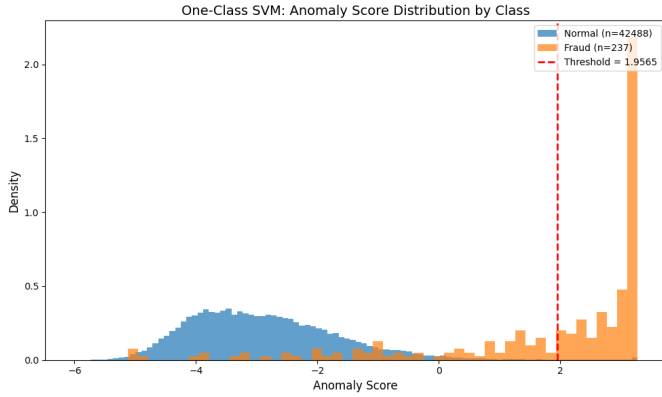


Fig. 6. Anomaly score distributions for One-Class SVM on the dev set (normal vs. fraud).

F1-score optimization on the dev set, the selected threshold is $\tau_{SVM} = 1.96$. Evaluation on the final test set yields:

- Precision: 0.413 (Appendix Fig. 12)
- Recall: 0.667 (Appendix Fig. 12)
- F1-score: 0.510

The OC-SVM correctly identifies 158 out of 237 fraud cases, with 225 false positives. This results in a false positive rate of 0.53% and a specificity of 0.995. The corresponding confusion matrix is shown in Fig. 7.

Threshold-independent metrics are:

- ROC-AUC: 0.946 (Appendix Fig. 11)
- PR-AUC: 0.384 (Appendix Fig. 12)

D. Comparative Analysis

A direct comparison between the two models shows complementary strengths. Overall, the Deep Autoencoder outperforms the One-Class SVM on four out of five major metrics:

- Higher precision and F1-score
- Superior ROC-AUC and PR-AUC

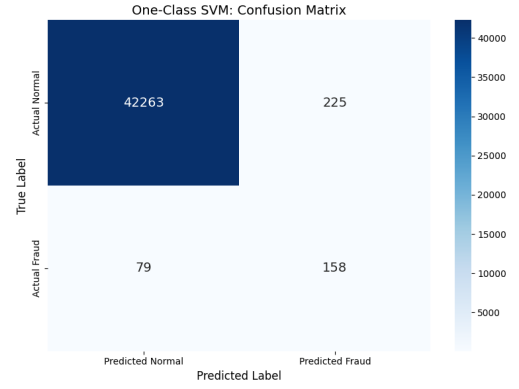


Fig. 7. Confusion matrix of the One-Class SVM on the final test set using τ_{SVM} .

- Lower false positive count

The OC-SVM achieves slightly higher recall (0.667 vs. 0.650), detecting four additional fraud cases, but at the cost of a higher number of false positives. Table I shows the distribution of the metrics among Deep Autoencoder and One-Class SVM.

TABLE I
DEEP AUTOENCODER VS ONE-CLASS SVM MODEL COMPARISON ON THE FINAL TEST SET.

Metric	Deep Autoencoder	One-Class SVM
Precision	0.4490	0.4125
Recall	0.6498	0.6667
F1-score	0.5310	0.5097
ROC-AUC	0.9606	0.9462
PR-AUC	0.4396	0.3844
True Positives (TP)	154	158
False Positives (FP)	189	225
True Negatives (TN)	42299	42263
False Negatives (FN)	83	79

Combined ROC and PR curves are shown in Appendix Figs.13 and 14, while a metric-level comparison is visualized in Appendix Fig. 15.

These results indicate that reconstruction-based anomaly detection provides a more favorable balance between fraud detection capability and alert volume under identical preprocessing and thresholding conditions.

V. CONCLUSION

This work investigated unsupervised anomaly detection for credit card fraud under extreme class imbalance and limited label availability. Two representative approaches—a Deep Autoencoder and a One-Class Support Vector Machine—were implemented and evaluated using a unified experimental pipeline.

Experimental results demonstrate that both models can successfully detect fraudulent transactions without using fraud labels during training. The Deep Autoencoder consistently achieves stronger overall performance, with higher precision, better F1-score, and superior ROC-AUC and PR-AUC values. The One-Class SVM, while achieving slightly higher recall, generates more false positives, making it less suitable in high-volume operational settings.

These findings suggest that reconstruction-based methods are particularly effective for high-dimensional financial transaction

data, where modeling normal behavior enables robust anomaly discrimination. Overall, this study confirms the viability of unsupervised learning for real-world fraud detection and identifies the Deep Autoencoder as a strong baseline for scenarios where labeled fraud data is scarce, delayed, or evolving.

APPENDIX

A. Individual Contributions

This project was carried out as a collaborative effort, with responsibilities distributed to ensure balanced participation across data preprocessing, model development, evaluation, and reporting.

Liza Berfin İnce

Liza Berfin İnce focused on Deep Autoencoder research and experimentation. She contributed to selecting the bottleneck architecture, tuning training hyperparameters, implementing early stopping and learning rate scheduling, and analyzing reconstruction error distributions. Liza also assisted in evaluating autoencoder performance using ROC-AUC, PR-AUC, and threshold-based metrics, as well as interpreting the corresponding visualizations.

Zeynep Deniz

Zeynep Deniz was primarily responsible for data handling and preprocessing. This included cleaning the dataset, removing duplicate transactions, performing feature scaling using normal-only statistics to prevent data leakage, and verifying data integrity through sanity checks. Zeynep also played a leading role in the design and implementation of the Deep Autoencoder model, contributing to architecture selection, training configuration, and reconstruction-error-based anomaly scoring.

Sait Kaçmaz

Sait Kaçmaz was primarily responsible for the One-Class SVM methodology. This included researching OC-SVM theory, configuring the RBF kernel and ν parameter, training the model on normal-only data, and defining anomaly scores from the decision function. He also led the threshold selection strategy for OC-SVM and performed detailed performance evaluation and comparison with the autoencoder.

Mohamad Bader

Mohamad Bader contributed to model development and comparative analysis with a focus on experimental rigor and evaluation design. He actively participated in validating the One-Class SVM pipeline, analyzing anomaly score behavior across different data splits, and interpreting performance under extreme class imbalance. Mohamad also played a key role in designing and implementing the unified evaluation protocol, generating comparative ROC and Precision–Recall visualizations, and synthesizing insights across both models to support a fair and consistent comparison.

B. Extra Materials

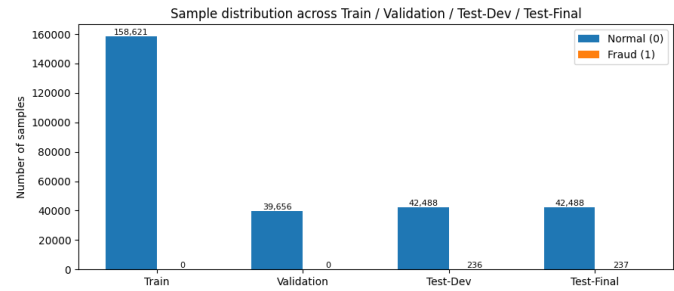


Fig. 8. Sample distribution across train/validation/test splits after preprocessing.

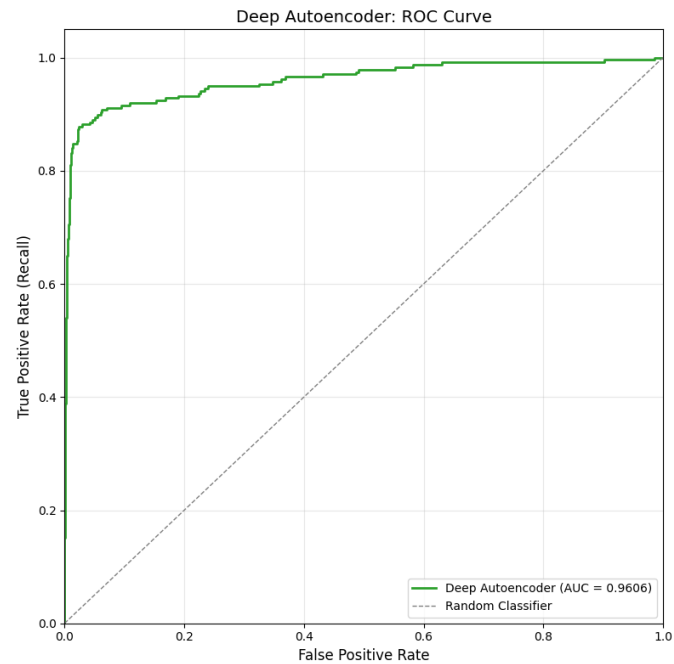


Fig. 9. ROC curve for the Deep Autoencoder on the final test set.

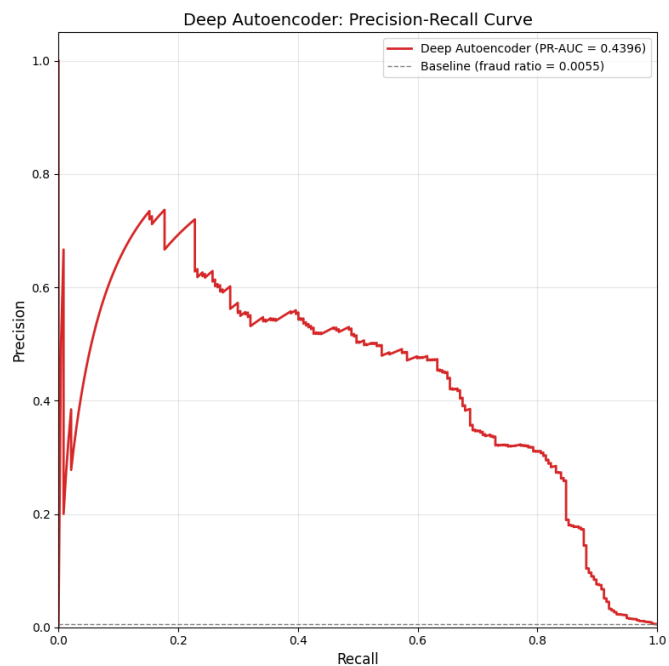


Fig. 10. Precision-Recall curve for the Deep Autoencoder on the final test set.

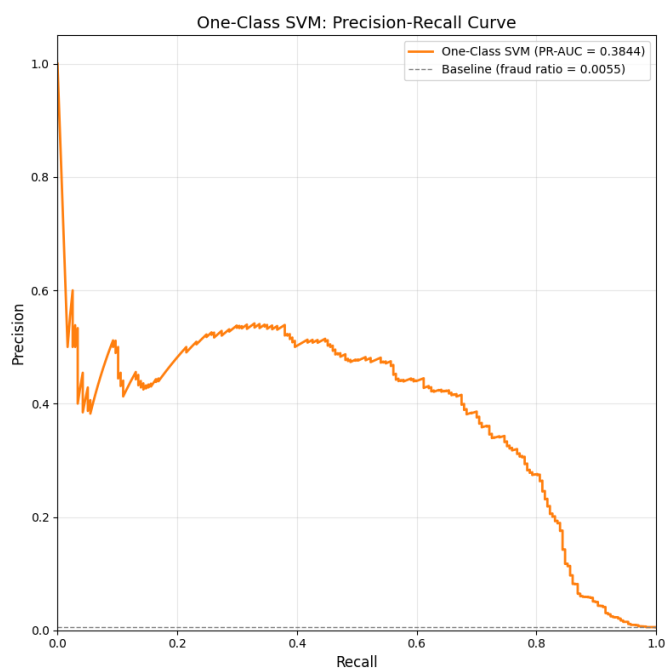


Fig. 12. Precision-Recall curve for One-Class SVM on the final test set.

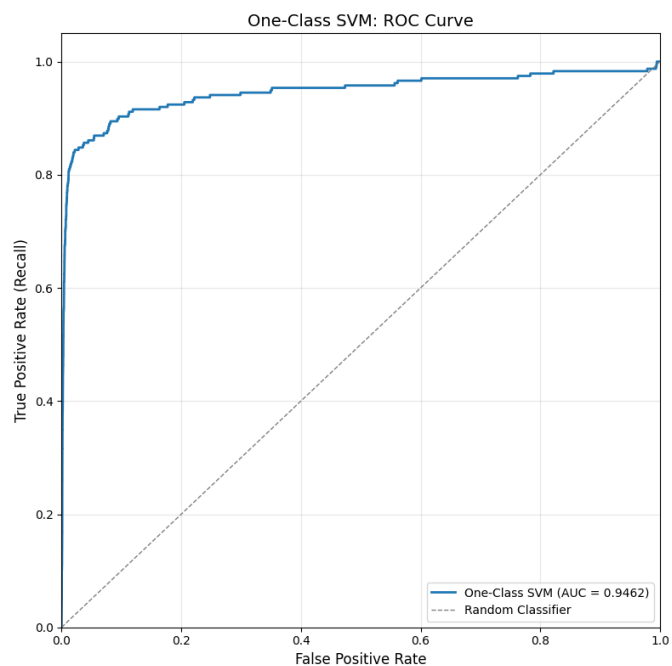


Fig. 11. ROC curve for One-Class SVM on the final test set.

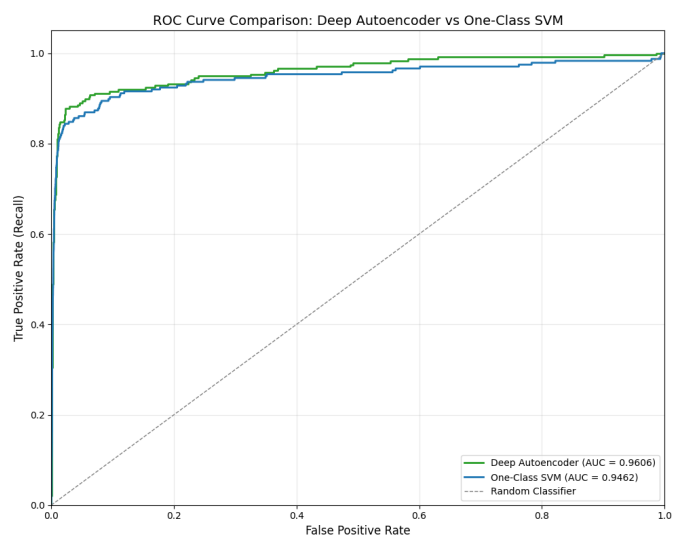


Fig. 13. Combined ROC curves comparing Deep Autoencoder and One-Class SVM on the final test set.

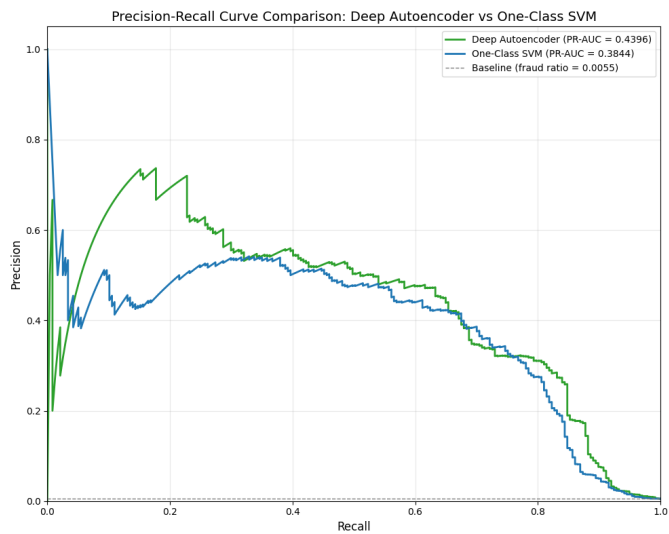


Fig. 14. Combined Precision-Recall curves comparing Deep Autoencoder and One-Class SVM on the final test set.

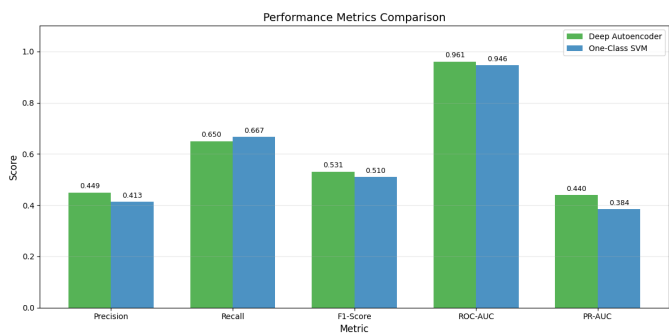


Fig. 15. Metric-level comparison between Deep Autoencoder and One-Class SVM on the final test set.

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